TRANSVECTIONS AND THE BRIDGE INDEX OF THIRTEEN-CROSSING KNOTS

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Abstract. KnotInfo lists the bridge index of 536 prime knots with thirteen crossings as either 3 or 4. We prove that 243 of them have bridge index 4 and meridional rank 4, so the meridional rank conjecture holds for these knots. The lower bound comes from representations of the knot group in $\mathrm{GL}_4(\mathbb{F}_q)$ that send a meridian to a transvection or a reflection and fix no nonzero vector. Each such representation is given explicitly and checked by a short program.

1. Introduction

Let $K \subset S^3$ be a knot, $G = \pi_1(S^3 \setminus K)$ its group and $m \in G$ a meridian. The *bridge index* $b(K)$ is the least number of local maxima of a height function on a knot isotopic to K. The *meridional rank* $\mu(K)$ is the least number of meridians, that is conjugates of $m^{\pm 1}$, that generate G. The overpasses of a b-bridge presentation give b meridians generating G, so

$$\mu(K) \leq b(K).$$

Cappell and Shaneson asked whether equality always holds [5, Problem 1.11]; this is the meridional rank conjecture.

Two lines of work make the conjecture checkable for a given knot. Blair, Kjuchukova, Velazquez and Villanueva defined the Wirtinger number $\omega(D)$ of a diagram D and proved that its minimum over the diagrams of K is $b(K)$ [3]. In particular

$$b(K) \leq \omega(D) \quad \text{for every diagram } D \text{ of } K. \tag{1}$$

Blair, Kjuchukova and Morrison bounded $\mu(K)$ from below by surjections of G onto Coxeter groups sending meridians to reflections, and so computed $b(K) = \mu(K)$ for 595,515 knots up to sixteen crossings [1]. KnotInfo [6] records the bridge index of thirteen-crossing knots from these bounds and from the plain sphere number [2]. In the version 2026.10.5 of the database, 536 knots with thirteen crossings have the entry $[3, 4]$: the bounds $3 \leq b(K) \leq 4$ are known and the value is not.

Theorem 1. *For each of the 243 knots K listed in Appendix A,*

$$b(K) = \mu(K) = 4.$$

In particular the meridional rank conjecture holds for these knots.

The upper bound is (1) applied to the KnotInfo diagram of K, whose Wirtinger number is 4. The lower bound is Lemma 2 below, applied to a representation $\rho\colon G \to \mathrm{GL}_4(\mathbb{F}_q)$ with $q \in \{3, 5\}$. Each ρ sends the meridian to a transvection of $\mathrm{Sp}_4(\mathbb{F}_q)$ or to a reflection of an orthogonal group $\mathrm{O}_4(\mathbb{F}_5)$. Reflections in Coxeter groups are pseudo-reflections too; the transvections, of order 3 and 5, are what settles most of these knots. Table 1 gives the counts.

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group	image of a meridian	knots
$\mathrm{O}_4(\mathbb{F}_5)$	reflections	15
$\mathrm{Sp}_4(\mathbb{F}_3)$	transvections	67
$\mathrm{Sp}_4(\mathbb{F}_5)$	transvections	161

TABLE 1. The certificates of Theorem 1, by the group used.

2. A lower bound from pseudo-reflections

A *pseudo-reflection* of a vector space V is an element $g \in \mathrm{GL}(V)$ with $\operatorname{rank}(g-1) = 1$. It fixes the hyperplane $\ker(g-1)$ pointwise.

Lemma 2. *Let V be a vector space of dimension n over a field and let $\rho\colon G \to \mathrm{GL}(V)$ be a homomorphism such that $\rho(m)$ is a pseudo-reflection. Let V^G be the subspace of vectors fixed by $\rho(G)$. Then*

$$\mu(K) \;\geq\; n - \dim V^G.$$

Proof. Let $m_1, \dots, m_k$ be meridians generating G, with $m_i = g_i m^{\epsilon_i} g_i^{-1}$ and $\epsilon_i = \pm 1$. Since $\rho(m)^{-1} - 1 = -\rho(m)^{-1}(\rho(m) - 1)$, each $\rho(m_i)$ is a pseudo-reflection, and $H_i = \ker(\rho(m_i) - 1)$ is a hyperplane. The intersection $H_1 \cap \cdots \cap H_k$ has dimension at least $n - k$ and is fixed by every $\rho(m_i)$, hence by $\rho(G)$. So $n - k \leq \dim V^G$. □

Corollary 3. *If $\dim V = 4$, $\rho(m)$ is a pseudo-reflection and $V^G = 0$, then $b(K) \geq \mu(K) \geq 4$.*

In a diagram with arcs $x_1, \dots, x_r$, the Wirtinger presentation makes such a ρ an assignment of matrices $X_i = \rho(x_i)$ to the arcs that satisfies $X_c = X_o^{\pm 1} X_a X_o^{\mp 1}$ at each crossing, where x_o is the overarc and x_a, x_c are the underarcs. Since the X_i generate $\rho(G)$,

$$V^G = \bigcap_{i=1}^{r} \ker(X_i - 1).$$

Both conditions are finite computations once the X_i are written down.

Transvections. Let ω be a nondegenerate alternating form on $V = \mathbb{F}_q^4$. For $v \neq 0$ the transvection $t_v(x) = x + \omega(x, v)\, v$ lies in $\mathrm{Sp}(V, \omega)$, $t_v - 1$ has rank one, $t_{-v} = t_v$, and $g\, t_v\, g^{-1} = t_{gv}$ for $g \in \mathrm{Sp}(V, \omega)$. A representation with $X_i = t_{v_i}$ is therefore the same thing as a coloring of the arcs by vectors with

$$v_c = \pm\, t_{v_o}^{\pm 1}(v_a)$$

at every crossing, and $\ker(X_i - 1) = v_i^{\perp}$. So $V^G = (\operatorname{span}\{v_i\})^{\perp}$, which is 0 exactly when the v_i span V.

Reflections. Let B be a nondegenerate symmetric bilinear form on $\mathbb{F}_q^4$, q odd. For $B(v, v) \neq 0$ the reflection $r_v(x) = x - 2\frac{B(x,v)}{B(v,v)}\, v$ satisfies $g\, r_v\, g^{-1} = r_{gv}$ for $g \in \mathrm{O}(B)$, and again $V^G = (\operatorname{span}\{v_i\})^{\perp}$.

3. The certificates

For each knot of Appendix A the file `data/certificates.json` contains:

- the PD code of K from KnotInfo, with edges numbered $1, \dots, 2n$ along the orientation and $[a, b, c, d]$ the edges at a crossing counterclockwise from the incoming underedge a;
- a prime q and a 4×4 matrix over $\mathbb{F}_q$ for every edge;
- four edges on distinct arcs, the seeds of a Wirtinger coloring.

The program `verify/verify.py` uses only the Python standard library. For every knot it checks:

(1) the matrices are constant along each arc, where an arc is a maximal run of edges between two undercrossings;
(2) at each crossing $[a, b, c, d]$ the relation $X_c = X_b^{-1} X_a X_b$ holds if the overstrand runs from b to d, and $X_c = X_b X_a X_b^{-1}$ if it runs from d to b (changing this convention at every crossing at once replaces K by its mirror image, which has the same b and μ);
(3) each $X_i - 1$ has rank 1, and the matrices $X_i - 1$ stacked together have rank 4, so that $V^G = 0$;
(4) starting from the four seed arcs, the coloring moves of [3] (if the overarc and one underarc at a crossing are colored, color the other underarc) color every arc, so $\omega(D) \leq 4$.

By Corollary 3 and (1), items (1)–(4) prove $b(K) = \mu(K) = 4$. With the option `--knotinfo` the program also checks that each PD code is the one in KnotInfo and that KnotInfo lists the bridge index as $[3, 4]$. All 243 certificates are checked in a few seconds. The program rejects altered certificates: a changed matrix entry, a seed set on three arcs, one crossing with its overstrand reversed, or identity matrices.

Remark 4. The certificates were found by a search, which plays no part in the proof. If the four seed vectors do not span V, every arc vector lies in their span, so a certificate must have seeds forming a basis. A coloring is then determined by the Gram matrix of the seeds: six entries of an alternating matrix, or six off-diagonal entries of a symmetric matrix with a fixed diagonal. The search runs over all such matrices and propagates the colors from the seeds through the crossings. The Wirtinger relations are then checked at the remaining crossings.

4. The knots that remain open

For the other 293 knots with entry $[3, 4]$ (Appendix B) we found neither bound. The exhaustive searches described above found no representation satisfying Corollary 3 with meridians sent to transvections of $\mathrm{Sp}_4(\mathbb{F}_q)$ for $q \in \{3, 4, 5, 7, 8, 9, 11\}$, to reflections of $\mathrm{O}_4(\mathbb{F}_q)$ for $q \in \{3, 5, 7\}$ (both isometry types), or to pseudo-reflections of $\mathrm{GL}_4(\mathbb{F}_2)$ and $\mathrm{GL}_4(\mathbb{F}_3)$ with eigenvalue 1 or -1. Random Reidemeister moves (SnapPy [4]) produced no diagram of Wirtinger number 3; 300 diagrams were tried for each knot. These knots stay open: a diagram of Wirtinger number 3 would give $b(K) = \mu(K) = 3$, and a representation as in Corollary 3 over another field, or a Coxeter quotient of rank 4, would give $b(K) = \mu(K) = 4$.

Data availability

The certificates, the verification program and the search programs are in the repository `https://github.com/creelie/conjecture-to-choose`. Release 1.0.0 is archived on Zenodo, doi:10.5281/zenodo.23521967.

References

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Appendix A. The knots of Theorem 1

$13a_{69}$, $13a_{70}$, $13a_{72}$, $13a_{73}$, $13a_{74}$, $13a_{76}$, $13a_{79}$, $13a_{96}$, $13a_{104}$, $13a_{106}$, $13a_{107}$, $13a_{108}$, $13a_{156}$, $13a_{157}$, $13a_{160}$, $13a_{171}$, $13a_{235}$, $13a_{343}$, $13a_{353}$, $13a_{354}$, $13a_{355}$, $13a_{358}$, $13a_{359}$, $13a_{360}$, $13a_{388}$, $13a_{390}$, $13a_{396}$, $13a_{399}$, $13a_{400}$, $13a_{402}$, $13a_{403}$, $13a_{408}$, $13a_{413}$, $13a_{414}$, $13a_{505}$, $13a_{524}$, $13a_{525}$, $13a_{526}$, $13a_{527}$, $13a_{528}$, $13a_{529}$, $13a_{535}$, $13a_{536}$, $13a_{541}$, $13a_{542}$, $13a_{543}$, $13a_{544}$, $13a_{545}$, $13a_{554}$, $13a_{555}$, $13a_{557}$, $13a_{558}$, $13a_{596}$, $13a_{597}$, $13a_{601}$, $13a_{621}$, $13a_{630}$, $13a_{631}$, $13a_{632}$, $13a_{636}$, $13a_{637}$, $13a_{657}$, $13a_{664}$, $13a_{665}$, $13a_{815}$, $13a_{867}$, $13a_{1392}$, $13a_{1538}$, $13a_{1995}$, $13a_{2006}$, $13a_{2008}$, $13a_{2613}$, $13a_{2614}$, $13a_{2615}$, $13a_{2616}$, $13a_{2617}$, $13a_{2619}$, $13a_{2620}$, $13a_{2800}$, $13a_{2801}$, $13a_{3038}$, $13a_{3040}$, $13a_{3107}$, $13a_{3144}$, $13a_{3146}$, $13a_{3179}$, $13a_{3257}$, $13a_{3333}$, $13a_{3334}$, $13a_{3343}$, $13a_{3344}$, $13a_{3376}$, $13a_{3377}$, $13a_{3378}$, $13a_{3380}$, $13a_{3381}$, $13n_{160}$, $13n_{161}$, $13n_{162}$, $13n_{163}$, $13n_{164}$, $13n_{165}$, $13n_{166}$, $13n_{167}$, $13n_{168}$, $13n_{169}$, $13n_{173}$, $13n_{174}$, $13n_{175}$, $13n_{176}$, $13n_{177}$, $13n_{178}$, $13n_{179}$, $13n_{180}$, $13n_{181}$, $13n_{182}$, $13n_{211}$, $13n_{429}$, $13n_{430}$, $13n_{431}$, $13n_{432}$, $13n_{433}$, $13n_{434}$, $13n_{435}$, $13n_{440}$, $13n_{441}$, $13n_{442}$, $13n_{449}$, $13n_{467}$, $13n_{601}$, $13n_{609}$, $13n_{610}$, $13n_{929}$, $13n_{930}$, $13n_{931}$, $13n_{932}$, $13n_{933}$, $13n_{957}$, $13n_{958}$, $13n_{959}$, $13n_{960}$, $13n_{961}$, $13n_{962}$, $13n_{963}$, $13n_{964}$, $13n_{965}$, $13n_{966}$, $13n_{967}$, $13n_{968}$, $13n_{977}$, $13n_{978}$, $13n_{979}$, $13n_{980}$, $13n_{981}$, $13n_{982}$, $13n_{983}$, $13n_{1042}$, $13n_{1058}$, $13n_{1059}$, $13n_{1060}$, $13n_{1061}$, $13n_{1062}$, $13n_{1063}$, $13n_{1064}$, $13n_{1072}$, $13n_{1073}$, $13n_{1074}$, $13n_{1075}$, $13n_{1076}$, $13n_{1077}$, $13n_{1078}$, $13n_{1082}$, $13n_{1083}$, $13n_{1084}$, $13n_{1103}$, $13n_{1104}$, $13n_{1105}$, $13n_{1106}$, $13n_{1239}$, $13n_{1280}$, $13n_{1281}$, $13n_{1282}$, $13n_{1283}$, $13n_{1284}$, $13n_{1285}$, $13n_{1286}$, $13n_{1287}$, $13n_{1288}$, $13n_{1289}$, $13n_{1290}$, $13n_{1291}$, $13n_{1292}$, $13n_{1293}$, $13n_{1294}$, $13n_{1295}$, $13n_{1305}$, $13n_{1306}$, $13n_{1307}$, $13n_{1308}$, $13n_{1309}$, $13n_{1310}$, $13n_{1311}$, $13n_{1312}$, $13n_{1313}$, $13n_{1314}$, $13n_{1315}$, $13n_{1316}$, $13n_{1317}$, $13n_{1318}$, $13n_{1319}$, $13n_{1320}$, $13n_{1321}$, $13n_{1322}$, $13n_{1323}$, $13n_{1324}$, $13n_{1332}$, $13n_{1333}$, $13n_{1334}$, $13n_{1363}$, $13n_{1377}$, $13n_{1378}$, $13n_{1379}$, $13n_{1380}$, $13n_{1381}$, $13n_{1385}$, $13n_{1386}$, $13n_{2933}$, $13n_{2934}$, $13n_{2935}$, $13n_{2936}$, $13n_{2937}$, $13n_{2954}$, $13n_{2955}$, $13n_{2956}$, $13n_{3390}$, $13n_{3505}$, $13n_{3723}$, $13n_{3725}$, $13n_{3726}$, $13n_{3727}$, $13n_{3763}$, $13n_{3838}$, $13n_{3915}$.

Appendix B. The knots with entry $[3, 4]$ that remain open

$13a_{129}$, $13a_{135}$, $13a_{136}$, $13a_{137}$, $13a_{194}$, $13a_{196}$, $13a_{197}$, $13a_{198}$, $13a_{233}$, $13a_{234}$, $13a_{239}$, $13a_{242}$, $13a_{243}$, $13a_{244}$, $13a_{253}$, $13a_{265}$, $13a_{268}$, $13a_{269}$, $13a_{270}$, $13a_{271}$, $13a_{272}$, $13a_{304}$, $13a_{309}$, $13a_{338}$, $13a_{362}$, $13a_{365}$, $13a_{366}$, $13a_{369}$, $13a_{373}$, $13a_{375}$, $13a_{387}$, $13a_{391}$, $13a_{392}$, $13a_{471}$, $13a_{483}$, $13a_{487}$, $13a_{516}$, $13a_{518}$, $13a_{530}$, $13a_{623}$, $13a_{629}$, $13a_{3145}$, $13a_{3147}$, $13a_{3148}$, $13a_{3149}$, $13a_{3167}$, $13a_{3168}$, $13a_{3169}$, $13a_{3170}$, $13a_{3188}$, $13a_{3189}$, $13a_{3190}$, $13a_{3191}$, $13a_{3204}$, $13a_{3205}$, $13a_{3207}$, $13a_{3208}$, $13a_{3209}$, $13a_{3210}$, $13a_{3213}$, $13a_{3216}$, $13a_{3219}$, $13a_{3220}$, $13a_{3221}$, $13a_{3222}$, $13a_{3223}$, $13a_{3232}$, $13a_{3234}$, $13a_{3248}$, $13a_{3259}$, $13a_{3261}$, $13a_{3262}$, $13a_{3263}$, $13a_{3264}$, $13a_{3266}$, $13a_{3274}$, $13a_{3275}$, $13a_{3276}$, $13a_{3291}$, $13a_{3292}$, $13a_{3293}$, $13a_{3294}$, $13a_{3295}$, $13a_{3298}$, $13a_{3301}$, $13a_{3302}$, $13a_{3303}$, $13a_{3317}$, $13a_{3320}$, $13a_{3323}$, $13a_{3330}$, $13a_{3331}$, $13a_{3335}$, $13a_{3345}$, $13a_{3346}$, $13a_{3347}$, $13a_{3348}$, $13a_{3351}$, $13a_{3354}$, $13a_{3355}$, $13a_{3356}$, $13a_{3363}$, $13a_{3365}$, $13a_{3368}$, $13a_{3374}$, $13a_{3375}$, $13a_{3388}$, $13a_{3390}$, $13a_{3393}$, $13a_{3394}$, $13a_{3395}$, $13a_{3396}$, $13a_{3399}$, $13a_{3400}$, $13a_{3401}$, $13a_{3402}$, $13a_{3403}$, $13a_{3404}$, $13a_{3405}$, $13a_{3407}$, $13a_{3409}$, $13a_{3410}$, $13a_{3411}$, $13a_{3412}$, $13a_{3413}$, $13a_{3415}$, $13a_{3417}$, $13a_{3418}$, $13a_{3811}$, $13a_{3812}$, $13n_{326}$, $13n_{329}$, $13n_{330}$, $13n_{341}$, $13n_{344}$, $13n_{345}$, $13n_{348}$, $13n_{351}$, $13n_{352}$, $13n_{355}$, $13n_{357}$, $13n_{358}$, $13n_{510}$, $13n_{513}$, $13n_{514}$, $13n_{518}$, $13n_{520}$, $13n_{521}$, $13n_{524}$, $13n_{525}$, $13n_{526}$, $13n_{529}$, $13n_{530}$, $13n_{531}$, $13n_{582}$, $13n_{590}$, $13n_{591}$, $13n_{598}$, $13n_{599}$, $13n_{600}$, $13n_{627}$, $13n_{632}$, $13n_{633}$, $13n_{644}$, $13n_{647}$, $13n_{648}$, $13n_{651}$, $13n_{652}$, $13n_{661}$, $13n_{662}$, $13n_{663}$, $13n_{694}$, $13n_{695}$, $13n_{696}$, $13n_{704}$, $13n_{705}$, $13n_{706}$, $13n_{707}$, $13n_{708}$, $13n_{709}$, $13n_{712}$, $13n_{713}$, $13n_{716}$, $13n_{717}$, $13n_{718}$, $13n_{834}$, $13n_{838}$, $13n_{839}$, $13n_{857}$, $13n_{926}$, $13n_{991}$, $13n_{992}$, $13n_{993}$, $13n_{1000}$, $13n_{1001}$, $13n_{1004}$, $13n_{1005}$, $13n_{1006}$, $13n_{1007}$, $13n_{1011}$, $13n_{1012}$, $13n_{1013}$, $13n_{1039}$, $13n_{1040}$, $13n_{1041}$, $13n_{1043}$, $13n_{1044}$, $13n_{1045}$, $13n_{1046}$, $13n_{1047}$, $13n_{1048}$, $13n_{1175}$, $13n_{1178}$, $13n_{1179}$, $13n_{1200}$, $13n_{1201}$, $13n_{1202}$, $13n_{1210}$, $13n_{1211}$, $13n_{1212}$, $13n_{1270}$, $13n_{1276}$, $13n_{1296}$, $13n_{1367}$, $13n_{1370}$, $13n_{1371}$, $13n_{1374}$, $13n_{1375}$, $13n_{1376}$, $13n_{3739}$, $13n_{3742}$, $13n_{3745}$, $13n_{3748}$, $13n_{3770}$, $13n_{3773}$, $13n_{3776}$, $13n_{3779}$, $13n_{3814}$, $13n_{3840}$, $13n_{3842}$, $13n_{3844}$, $13n_{3854}$, $13n_{3855}$, $13n_{3856}$, $13n_{3870}$, $13n_{3871}$, $13n_{3872}$, $13n_{3873}$, $13n_{3874}$, $13n_{3897}$, $13n_{3898}$, $13n_{3899}$, $13n_{3900}$, $13n_{3901}$, $13n_{3903}$, $13n_{3904}$, $13n_{3905}$, $13n_{3911}$, $13n_{3914}$, $13n_{3916}$, $13n_{3918}$, $13n_{3919}$, $13n_{3920}$, $13n_{3921}$, $13n_{3922}$, $13n_{3924}$, $13n_{3925}$, $13n_{3928}$, $13n_{3929}$, $13n_{3931}$, $13n_{3932}$, $13n_{3933}$, $13n_{3934}$, $13n_{3935}$, $13n_{3936}$, $13n_{3937}$, $13n_{3938}$, $13n_{3939}$, $13n_{3940}$, $13n_{3941}$, $13n_{4587}$, $13n_{4589}$, $13n_{4633}$, $13n_{4634}$, $13n_{4637}$, $13n_{4638}$, $13n_{4639}$, $13n_{4640}$, $13n_{4641}$, $13n_{4642}$, $13n_{4643}$, $13n_{4644}$, $13n_{4645}$.

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